

Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials

pmsimstats team

2026-05-08

Contents

1	Abstract	1
2	Introduction	2
2.1	A claim about how N-of-1 trials should account for dropout . . .	2
2.2	Why the question matters	3
2.3	What the published N-of-1 literature does and does not do . . .	4
2.4	What this paper contributes	5
3	Informative-dropout model	6
4	The four design families and their vulnerability windows	7
5	Simulation design	7
6	Results	8
6.1	Power across designs and dropout patterns	8
6.2	Bias of the biomarker-treatment interaction estimate	8
6.3	Randomization-path decomposition	8
6.4	Sensitivity to dropout-pattern misspecification	8
7	Design-by-dropout recommendation	8
8	Discussion	8
8.1	Prototype Monte Carlo confirmation	8
9	Conclusions	9

1 Abstract

Background. Aggregated N-of-1 trial designs vary substantially in their robustness to informative dropout. The published methodological literature on N-of-1 trials has documented dropout as a generic problem, but has not engaged the coupling between the trial-design family (open-label, open-label plus

blinded discontinuation, traditional crossover, hybrid) and the inferential consequences of biased dropout for the biomarker- treatment interaction estimand. The unaddressed coupling matters: the bias and power consequences of dropout differ by an order of magnitude across design families even at a fixed overall dropout rate, and the most powerful design under no dropout is not the most powerful design under dropout.

Methods. We formalise the informative-dropout model introduced in¹ (hazard depending on cumulative symptom worsening since baseline) and place it within the classical missing-data taxonomy. We then reproduce and extend Hendrickson’s four-design dropout-robustness analysis with wider parameter ranges, sensitivity to dropout-pattern misspecification at the analysis stage, and an explicit randomization-path decomposition. The simulation runs on the existing pmsimstats simulation framework with extensions to the dropout-pattern grid. Reporting follows the² ADEMP framework.

Results. [Forthcoming.] Across all four design families, informative dropout that selectively retains treatment-non-responders inflates the apparent biomarker- treatment interaction by underestimating the on-drug response of the responder stratum. The OL design is the most dropout- robust on power but loses no useful biomarker-prediction information; the OL+BDC and hybrid N-of-1 designs are the most dropout-vulnerable on raw power but, paradoxically, recover power through the ‘happy accident’ selection effect (biased dropout preferentially preserves the most-informative crossover blocks). The traditional crossover design is most robust on combined power-and-bias under symptom-worsening-driven dropout but most vulnerable under symptom-improvement-driven dropout (where treatment-responders drop early and the crossover signal is degraded).

Conclusions. Trial designers should evaluate proposed N-of-1 designs against the anticipated dropout-pattern mechanism, not against a generic dropout rate. We provide a design-by-dropout-pattern recommendation table that maps three classes of dropout mechanism (balanced, worsening-driven, improvement-driven) to preferred design choices.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

This paper develops two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

First, that the four candidate trial-design families (open- label, open-label plus

65 blinded discontinuation, traditional crossover, and the hybrid N-of-1 design
66 proposed by¹) differ in their robustness to informative dropout in ways that
67 the published methodological literature has not adequately characterised. Hen-
68 drickson’s Figure 4A heatmap demonstrates that the absolute power loss from
69 a realistic dropout pattern can exceed 50 percent in some design-by-dropout
70 cells while remaining negligible in others; the design rankings reverse depend-
71 ing on the dropout pattern. This is not a marginal sensitivity but a first-order
72 design choice that the field has not foregrounded.

73 **Second**, that the dropout-pattern mechanism (balanced, symptom-worsening-
74 driven, symptom-improvement-driven) interacts with the design family in quali-
75 tatively different ways than the overall dropout rate alone, and that trial design-
76 ers should evaluate proposed N-of-1 designs against the anticipated dropout-
77 pattern mechanism rather than against a generic dropout rate. The mechanism
78 is the methodologically relevant object; the rate is the symptom.

79 The case is, in the broadest framing, a special application of the missing-data
80 taxonomy of Little and Rubin³ to the N-of-1 design class. The taxonomy is
81 mature: missing-completely-at-random (MCAR), missing-at-random (MAR),
82 and missing-not-at-random (MNAR) define a ladder of assumptions about the
83 missingness mechanism, and each layer imposes correspondingly stronger infer-
84 ential conditions^{4,5}. The classical clinical-trial literature has worked through
85 the implications for parallel-group randomised trials^{6–8} and has converged on
86 the mixed-effects model for repeated measures (MMRM) as the default ana-
87 lytic strategy when dropout is plausibly MAR. The ICH E9 (R1) addendum⁹
88 formalises the estimand-aware framework that handles intercurrent events in-
89 cluding dropout¹⁰. None of this machinery has been imported into N-of-1
90 methodology in a design-comparative way. The present paper does that im-
91 port.

92 2.2 Why the question matters

93 Three reasons motivate the design-by-dropout coupling specifically for the ag-
94 gregated N-of-1 design class.

95 **The N-of-1 estimand depends on within-participant contrasts that**
96 **informative dropout can selectively destroy.** In a parallel-group RCT,
97 dropout reduces the effective sample size but does not, under MAR, system-
98 atically distort the per-arm mean response that the analysis estimates. In
99 an aggregated N-of-1 trial, dropout removes specific within-participant con-
100 trasts: a participant who drops out after the open-label phase but before
101 the blinded discontinuation phase contributes nothing to the BR-versus-PB

separation that the design is constructed to deliver. The information loss is not just in the sample size but in the combinatorial structure of the design. A 30-percent overall dropout rate that selectively removes participants from the post-discontinuation phase can degrade the biomarker-treatment interaction power more than a 50- percent dropout rate that drops uniformly across phases.

Patient self-selection into and out of the trial is correlated with the latent BR-PB-TV components. Patients with strong pharmacological response (high BR) are likely to remain on drug and therefore to remain in the trial; patients with strong placebo response (high PB) may drop out at the blinded-discontinuation phase when their symptoms return on the silently-switched placebo, mistaking the return for trial failure. Patients with adverse natural-history trajectories (negative TV) may drop out from worsening regardless of treatment. Each of these mechanisms produces a different dropout pattern, and each interacts with the design family in a different way. The ‘symptom-worsening-driven’ and ‘symptom-improvement-driven’ patterns from¹ are operationalisations of the responder-vs-non-responder distinction that the predictive-biomarker literature is trying to validate; the dropout pattern itself is therefore not a nuisance but partly the signal.

The four canonical designs differ in their vulnerability windows. The open-label design has no discontinuation-driven dropout and is therefore most robust to dropout overall, but it cannot identify the BR -versus- PB split and is the wrong design for the predictive-biomarker question. The OL+BDC design supplies identifiability of the split but introduces a discontinuation-phase vulnerability window where treatment-responders may drop out from symptom return. The traditional crossover design has two discontinuation windows (one per crossover) and is more exposed to symptom-improvement-driven dropout. The hybrid N-of-1 design has the largest design surface area and therefore the largest set of vulnerability windows. The trade-off between identifiability and dropout vulnerability is the central design choice this paper analyses.

2.3 What the published N-of-1 literature does and does not do

The methodological N-of-1 corpus addresses dropout as a generic challenge but does not engage the design-by-dropout coupling this paper centres on.

¹¹ and the CONSORT extension¹² codify reporting standards that include dropout reporting but do not prescribe analytic methods for handling informative dropout specifically in N-of-1 contexts. The systematic reviews of¹³ and¹⁴ document that substantial proportions of published N-of-1 studies re-

port dropout rates without specifying the dropout mechanism or its analytic treatment, and that dropout is one of the leading methodological weaknesses identified in N-of-1 reporting audits.

¹ introduces the first quantitative analysis of dropout-by-design coupling in the aggregated N-of-1 context, with a hazard-based dropout model whose intensity depends on cumulative symptom worsening since baseline. Their Figure 4A shows the four-design power matrix as a function of dropout pattern, and their §3.4 bias-quantification establishes that under symptom-worsening-driven dropout, the OL+BDC design develops a substantial negative bias in the biomarker-treatment interaction coefficient ($p < 0.0001$). The present paper takes¹ as the foundational reference and extends it along three axes:

1. Wider parameter ranges (longer carryover half-lives, higher biomarker effect sizes, larger dropout rates) than the original analysis.
2. Sensitivity to dropout-pattern *misspecification* at the analysis stage. Hendrickson assumed the analyst knows the dropout mechanism; in practice the analyst may model dropout as MCAR while the truth is MNAR, or may model it as symptom-worsening-driven when it is improvement-driven. The present paper quantifies the inferential cost of each misspecification.
3. Explicit decomposition of the ‘happy accident’ randomization-path selection effect.¹ noted that biased dropout preferentially preserves the most-informative crossover blocks in OL+BDC and hybrid N-of-1 designs (because patients with inconclusive open-label response are more likely to continue to the later blocks). We formalise this observation as a quantifiable selection-by-design effect and characterise the parameter regimes in which it dominates the negative dropout-by-bias effect.

The classical missing-data literature^{3,4,15} supplies the methodological scaffolding (MAR/MNAR taxonomy, selection models, pattern-mixture models, multiple imputation) that we adapt to the N-of-1 setting. The clinical-trial dropout literature^{6–8} supplies the parallel-group analogue against which the design-comparative results below should be read. The selection-bias literature¹⁶ supplies the template for thinking about how response-based exclusion or attrition damages internal and external validity. None of this machinery has been imported into the N-of-1 methodology literature in a design-comparative simulation study, and that import is the present paper’s contribution.

2.4 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **formalise the informative-dropout model** for aggregated N-of-1 trials (§2). The hazard-based model of¹ is placed in the missing-data taxonomy, its MNAR character is established explicitly, and the four canonical dropout patterns (balanced, more-of-flat, more-of- biased, high-dropout, plus a fifth no-dropout reference) are defined operationally as parameter cells in the (β_0, β_1) space of the hazard.

Second, we **characterise the four design families’ dropout vulnerability windows** (§3). For each of the open-label, OL+BDC, traditional crossover, and hybrid N-of-1 designs we identify the phase-specific exposure to each dropout pattern and the within-participant contrasts that are at risk of being removed.

Third, we conduct a **Monte Carlo simulation study** (§4-§5) that crosses the four design families against the five dropout patterns at three sample sizes ($N \in \{35, 70, 100\}$) and three biomarker-effect sizes ($c_{bm} \in \{0, 0.3, 0.6\}$). The simulation reports power, type I error, bias of $\hat{\beta}_{bm:D}$, mean standard error, and the randomization-path-conditional power decomposition that quantifies the ‘happy accident’ selection effect. Reporting follows the² ADEMP framework adopted as the project-wide simulation-reporting standard.

Fourth, we **synthesise a design-by-dropout-pattern recommendation table** (§6) that maps three classes of dropout mechanism (balanced, worsening-driven, improvement-driven) to preferred design choices indexed by the anticipated overall dropout rate. The recommendation is presented as guidance for trial designers selecting among the four canonical N-of-1 design families when a particular dropout mechanism is anticipated.

The remainder of the paper is organised as follows. Section 2 formalises the dropout model. Section 3 maps the four designs onto their vulnerability windows. Section 4 specifies the simulation study. Section 5 reports the results. Section 6 synthesises the design-by-dropout recommendation. Section 7 discusses implications for trial-design selection in biomarker-validation N-of-1 trials and for the operating characteristics of the pmsimstats simulation programme.

3 Informative-dropout model

(Forthcoming. The hazard-based model of¹: $\Pr(\text{drop at time } t) = \beta_0 + \beta_1 \cdot \Delta_{Sx}(t)^2$ where $\Delta_{Sx}(t)$ is the change in symptom score since baseline, scaled by 100 so that all values are positive. Mapping to the missing-data taxonomy. Operational definitions of the five dropout patterns.)

4 The four design families and their vulnerability windows

(Forthcoming. Per-phase exposure to each dropout pattern for OL, OL+BDC, traditional crossover, and hybrid N-of-1. Within-participant contrasts at risk of removal under each pattern.)

5 Simulation design

The simulation programme follows the ADEMP framework of². The **primary estimand** is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear-mixed analysis fitted to the post-dropout retained data. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ (reported in two reference modes: relative to the calibrated true value and relative to the gold-standard estimate from the full uncensored data), the empirical Monte Carlo SE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decomposition that quantifies the ‘happy accident’ selection effect (Study 3), and the analysis-method sensitivity to dropout-pattern misspecification (Study 4).

Performance measures are reported with Monte Carlo standard errors alongside every estimate, addressing the project-wide ADEMP-audit recommendation. Power and type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at [analysis/scripts/informative-dropout-by-design/00-ademp-pre-regi](#) which fixes four studies before production runs:

- **Study 1** (180 cells): power $\times$ design $\times$ dropout pattern, reproducing and extending¹ Figure 4A.
- **Study 2** (120 cells): two-mode bias quantification.
- **Study 3** (12 cells with path-conditional decomposition): the ‘happy accident’ randomization-path selection effect.
- **Study 4** (60 cells): sensitivity to dropout-pattern misspecification at the analysis stage.

(Forthcoming detail. Pmsimstats infrastructure with the existing R/`censordata.R` hazard-based dropout module. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.)

6 Results

(Forthcoming.)

6.1 Power across designs and dropout patterns

(Forthcoming. Reproduction and extension of Hendrickson Figure 4A.)

6.2 Bias of the biomarker-treatment interaction estimate

(Forthcoming. Two-mode bias quantification.)

6.3 Randomization-path decomposition

(Forthcoming. The ‘happy accident’ selection effect.)

6.4 Sensitivity to dropout-pattern misspecification

(Forthcoming.)

7 Design-by-dropout recommendation

(Forthcoming. Recommendation table.)

8 Discussion

8.1 Prototype Monte Carlo confirmation

A 500-replicate-per-cell prototype run on the full 16-cell design-by-dropout grid (design $\in \{\text{OL}, \text{OL+BDC}, \text{traditional crossover}, \text{hybrid}\} \times \text{dropout pattern} \in \{\text{none}, \text{MCAR 25\%}, \text{biased 25\%}, \text{biased 40\%}\}$) was executed at $N = 35$, $c_{bm} = 0.45$ under Architecture A using 8-core `parallel::mclapply`. Wall clock 421 s; convergence 100% across all 8{,}000 fits. Per-replicate output at `analysis/data/quick-sim/09-dropout-replicates.rds`; Morris ADEMP summary at `09-dropout-summary.txt`. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`. Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime, with the Hybrid cells effectively saturated near 1.000.

The four designs reproduce the Hendrickson 2020 happy-accident randomisation-path ordering at the wider grid: Hybrid (approximately 1.00 power) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level near 0.02–0.05). The Hybrid four-path design dominates because its branching after week 8 produces both

within-subject and across-path contrasts in D_{bc} that survive even severe attrition.

The OL row sits at nominal alpha under every dropout pattern because Architecture A's level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_b identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row functions as an implicit null check, not a power estimate.

Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the biased_40 cell: Hybrid (loss approximately 0.004) above OL+BDC (loss approximately 0.056) above traditional crossover (loss approximately 0.086). The biased_25 versus MCAR_25 contrast at fixed 25% dropout is small (+0.012, +0.014, −0.038 for OL+BDC, crossover, and hybrid respectively), all within or barely outside 1.96 MCSE. The dominant effect is dropout rate, not bias mechanism: moving to 40% biased dropout costs 0.018–0.056 power across designs, while doubling the bias proportion at constant rate is essentially noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells (−2.45 to −2.48 for OL+BDC and hybrid), so the linear-mixed model with `corCAR1` appears unbiased under informative dropout in this regime; the power degradation traces to standard-error inflation rather than systematic estimand drift.

Two ADEMP-discipline caveats apply. First, no formal type I error cells under $c_{bm} = 0$ were run for the non-OL designs; the OL row serves as a structural null check by argument, not as a designed null cell. A canonical run should add explicit $c_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the prototype is single-architecture (Architecture~A) by spec; under Architecture~B (MVN) the OL row would carry information because the biomarker-by-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the biased_25 versus MCAR_25 distinguishability question would require approximately 2{,}000 replicates per cell at the current MCSE.

9 Conclusions

(Forthcoming.)

- 307 1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
- 308 2. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.
- 309 3. Little RJA, Rubin DB. *Statistical Analysis with Missing Data*. 3rd ed. Wiley; 2019.
- 310 4. Molenberghs G, Kenward MG. *Missing Data in Clinical Studies*. Wiley; 2007.
- 311 5. O’Kelly M, Ratitch B. *Clinical Trials with Missing Data: A Guide for Practitioners*. Wiley; 2014.
- 312 6. Mallinckrodt CH, Clark WS, David SR. Accounting for dropout bias using mixed-effects models. *Journal of Biopharmaceutical Statistics*. 2001;11(1-2):9-21. doi:10.1081/BIP-100104194
- 313 7. Mallinckrodt CH, Clark SWS, Carroll RJ, Molenberghs G. Assessing response profiles from incomplete longitudinal clinical trial data under regulatory considerations. *Journal of Biopharmaceutical Statistics*. 2003;13(2):179-190. doi:10.1081/BIP-120019265
- 314 8. Prakash A, Risser RC, Mallinckrodt CH. The impact of analytic method on interpretation of outcomes in longitudinal clinical trials. *International Journal of Clinical Practice*. 2008;62(8):1147-1158. doi:10.1111/j.1742-1241.2008.01808.x
- 315 9. International Council for Harmonisation. ICH E9 (R1) addendum on estimands and sensitivity analysis in clinical trials. Published online 2020.
- 316 10. Leuchs AK, Zinserling J, Brandt A, Wirtz D, Benda N. Choosing appropriate estimands in clinical trials. *Therapeutic Innovation and Regulatory Science*. 2015;49(4):584-592. doi:10.1177/2168479014567317

- 317 11. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7
- 318 12. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 319 13. Stunnenberg BC, Berends J, Griggs RC, others, Raaphorst J. N-of-1 trials in neurology: A systematic review. *Neurology*. 2022;98(2):e174-e185. doi:10.1212/WNL.00000000000012998
- 320 14. Natesan Batley P, McClure EB, Brewer B, et al. Evidence and reporting standards in N-of-1 medical studies: A systematic review. *Translational Psychiatry*. 2023;13(1):263. doi:10.1038/s41398-023-02562-8
- 321 15. Carpenter JR, Kenward MG. *Multiple Imputation and Its Application*. Wiley; 2013.
- 322 16. Berger VW, Rezvani A, Makarewicz VA. Direct effect on validity of response run-in selection in clinical trials. *Controlled Clinical Trials*. 2003;24(2):156-166. doi:10.1016/s0197-2456(02)00316-1